

MATH 127 Calculus for the Sciences

Lecture 23



Today's lecture

Last time

How to solve indeterminate forms $\frac{\infty}{\infty}, \frac{0}{0}$.

This time

Course note coverage Section 3.2.4 and 3.2.5

Indeterminate product and indeterminate powers.



Happy Halloween

I have chocolate for you. Here is a picture of me visiting the Lindt chocolate factory to buy you this great chocolate.



Lindt  HOME OF CHOCOLATE



Recap

Determinate form	evaluation
$\frac{1}{0^+}, \frac{\infty}{0^+}, \infty + \infty, \infty^\infty$	∞
$\frac{1}{0^-}, \frac{\infty}{0^-}$	$-\infty$
$\frac{0}{\infty}, \frac{1}{\infty}, 0^\infty$	0
Indeterminate form	tricks
$\frac{\infty}{\infty}, \frac{0}{0}$	L'Hospital's rule simplify multiply conjugate
$0 \cdot \infty$ (new!)	will see today
$0^0, \infty^0, 1^\infty$ (new!)	will see today



Indeterminate product

Question What do we do with a indeterminate form like $0 \cdot \infty$?

We could rewrite

$$0 \cdot \infty = \frac{0}{(1/\infty)} = \frac{0}{0}$$

And we know how to deal with $\frac{0}{0}$ from before!

Alternatively...

$$0 \cdot \infty = \frac{\infty}{(1/0)} = \frac{\infty}{\infty}$$

And we know how to deal with $\frac{\infty}{\infty}$ from before!



Example: Indeterminate product

Example

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x} \right) \cdot \sin(x)$$

The limit of $\frac{1}{x}$ as $x \rightarrow 0^+$ is

The limit of $\sin(x)$ as $x \rightarrow 0^+$ is

So this limit has indeterminate form

But we could rewrite the limit as a quotient

$$\lim_{x \rightarrow 0^+} \frac{\sin(x)}{\frac{1}{x}}$$

This has indeterminate form

which can be solved using rule.



Example: Indeterminate product

Example

$$\lim_{x \rightarrow \infty} x \cdot \sin\left(\frac{1}{x}\right)$$

The limit of x as $x \rightarrow \infty$ is

11

The limit of $\sin\left(\frac{1}{x}\right)$ as $x \rightarrow \infty$ is

So this limit has indeterminate form

11/11/2019

But we could rewrite the limit as a quotient

$$\lim_{x \rightarrow \infty} \frac{x}{1/\sin\left(\frac{1}{x}\right)}, \quad \text{or} \quad \lim_{x \rightarrow \infty} \frac{\sin\left(\frac{1}{x}\right)}{1/x}$$

These have indeterminate forms

and

respectively... but which one should we use?



Example: Indeterminate product

Example

$$\lim_{x \rightarrow \infty} \frac{x}{1/\sin\left(\frac{1}{x}\right)}, \quad \text{or} \quad \lim_{x \rightarrow \infty} \frac{\sin\left(\frac{1}{x}\right)}{1/x}$$

These have indeterminate forms

$$\frac{\infty}{\infty} \text{ and } \frac{0}{0}$$

respectively... but which one should we use?

We are planning to apply rule. So we need to take of the denominator and numerator. Which one is easier?



Example: Indeterminate product

Example

$$\lim_{x \rightarrow \infty} \frac{x}{1/\sin\left(\frac{1}{x}\right)}, \quad \text{or} \quad \lim_{x \rightarrow \infty} \frac{\sin\left(\frac{1}{x}\right)}{1/x}$$

The **second one** is easier to apply L'Hospital's rule, because who on earth would want to take the derivative of $1/\sin(1/x)$?



Indeterminate power

Question What do we do with a indeterminate form like $0^0, \infty^0, 1^\infty$?

Did you forget what I said on Sept 24? Always use base e !

Rewrite

$$0^0 = e^{0 \ln 0} = e^{-0 \cdot \infty}$$

Similarly

$$\infty^0 = e^{0 \cdot \ln \infty} = e^{0 \cdot \infty}$$

and

$$1^\infty = e^{\infty \cdot \ln 1} = e^{\infty \cdot 0}$$

Now we just have to solve for the $0 \cdot \infty$ thing (which we just saw how to do) and take e to the power of it.



Example: indeterminate power

Question Evaluate $\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$.



Example: indeterminate power

Question Evaluate $\lim_{x \rightarrow 0} \sin(x)^{\tan(x)}$.



Recap

Determinate form	evaluation
$\frac{1}{0^+}, \frac{\infty}{0^+}, \infty + \infty, \infty^\infty$	∞
$\frac{1}{0^-}, \frac{\infty}{0^-}$	$-\infty$
$\frac{0}{\infty}, \frac{1}{\infty}, 0^\infty$	0
Indeterminate form	tricks
$\frac{\infty}{\infty}, \frac{0}{0}$	L'Hospital's rule simplify multiply conjugate
$0 \cdot \infty$	rewrite as a quotient and apply above
$0^0, \infty^0, 1^\infty$	rewrite in base e and apply above